

KARLSTAD AP, Sweden

WMO: 024180

Lat: 59.445N

Lon: 13.337E

Elev: 352

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.2	(c) 6.5	(d) -2.6	(e) 4.8	(f) 3.0	(g) 2.3	(h) 6.3	(i) 9.3	(j) 22.8	(k) 37.8	(l) 20.6	(m) 37.3	(n) 3.8	(o) 0	(p) 0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	79.2	63.6	76.4	62.6	73.2	61.2	66.6	74.6	64.7	72.4	63.0	70.3	8.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 64.0	(b) 90.4	(c) 69.4	(d) 62.0	(e) 84.2	(f) 67.5	(g) 60.1	(h) 78.6	(i) 66.1	(j) 31.3	(k) 74.5	(l) 29.9	(m) 72.5	(n) 28.7	(o) 70.1	(p) 73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	16.8	15.1	-4.0	83.2	6.1	3.4	-8.4	85.6	-11.9	87.7	-15.3	89.6	-19.8	92.1
			-4.4	68.5	6.0	2.3	-8.7	70.2	-12.2	71.6	-15.6	72.8	-19.9	74.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	26.6	27.2	32.8	41.6	50.3	57.5	62.4	60.4	53.0	43.5	35.6	29.0
	DBStd	14.46	9.36	9.02	7.26	6.12	6.54	4.85	4.89	4.83	5.30	6.55	7.22	10.00
	HDD50	3572	727	639	532	262	76	2	0	1	32	216	434	651
	HDD65	7937	1192	1059	997	702	458	231	113	158	362	666	883	1116
	CDD50	1161	0	0	0	10	84	227	383	322	120	15	0	0
	CDD65	52	0	0	0	0	1	6	31	14	0	0	0	0
	CDH74	529	0	0	0	0	22	90	313	104	0	0	0	0
	CDH80	66	0	0	0	0	1	10	52	3	0	0	0	0
Wind	WSAvg	7.1	7.3	7.3	7.3	7.2	7.3	7.0	6.5	6.4	7.1	7.2	7.5	7.1
Precipitation	PrecAvg	27.9	2.2	1.6	1.5	1.9	2.1	2.7	2.5	3.1	2.7	3.0	2.8	2.2
	PrecMax	38.6	4.7	3.2	3.9	4.7	4.0	4.9	4.8	5.9	5.5	7.2	7.0	4.2
	PrecMin	20.6	0.4	0.6	0.2	0.5	0.3	0.4	0.4	1.2	0.7	1.0	1.1	0.7
	PrecStd	4.4	1.2	0.7	0.8	1.1	1.1	1.3	1.1	1.5	1.3	1.6	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	46.6	56.9	67.8	77.1	81.1	84.9	79.9	71.5	60.0	52.9	49.5
		MCWB	42.2	42.3	46.1	53.2	59.7	62.7	65.1	64.5	61.8	54.7	50.4	47.1
	2%	DB	42.1	43.1	50.3	62.3	72.0	76.6	80.8	77.0	67.5	57.0	50.2	45.8
		MCWB	40.3	39.7	42.6	50.3	56.7	60.6	64.2	63.9	58.9	53.5	48.2	44.4
	5%	DB	40.0	40.7	46.3	57.3	67.9	72.0	77.4	73.8	64.4	55.2	48.0	43.5
		MCWB	38.4	38.2	40.2	47.4	55.0	58.6	63.4	63.1	57.7	52.5	46.4	42.0
	10%	DB	37.9	38.7	42.8	53.3	63.4	68.4	73.9	70.5	61.8	53.2	45.9	41.3
		MCWB	36.5	36.8	37.9	44.8	53.0	57.0	62.0	61.6	56.2	50.8	44.5	40.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	43.1	47.3	55.6	62.5	65.3	69.0	68.9	63.6	56.7	51.3	47.3
		MCDB	43.9	45.4	54.8	63.9	72.3	75.6	77.3	74.8	67.9	58.1	52.5	48.6
	2%	WB	40.6	40.5	43.5	50.8	59.1	62.7	67.0	66.5	60.7	54.7	48.6	44.7
		MCDB	41.4	42.0	48.8	59.5	67.0	72.3	76.3	72.7	65.6	56.1	49.7	45.4
	5%	WB	38.6	38.7	41.1	48.6	56.4	60.5	65.2	64.6	58.8	52.9	46.4	42.1
		MCDB	39.6	40.1	45.1	55.9	65.1	69.3	74.0	70.9	62.9	54.7	47.5	43.1
	10%	WB	36.8	36.8	38.9	46.1	54.0	58.7	63.5	62.9	57.1	51.2	44.5	40.0
		MCDB	37.8	38.3	41.9	51.4	62.6	66.2	71.5	68.8	60.9	52.9	45.5	41.0
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.4	13.3	16.9	17.8	17.0	17.0	16.1	14.5	11.3	8.4	8.9
		MCDBR	8.9	10.6	18.3	24.1	25.2	23.8	23.6	21.1	18.2	11.9	10.4	8.5
		MCWBR	8.6	8.4	12.2	13.8	12.5	11.2	10.2	10.5	11.0	9.0	9.7	8.4
	5% WB	MCDBR	8.4	9.6	16.6	21.0	21.3	20.3	20.2	17.9	14.9	10.1	9.9	8.6
		MCWBR	8.3	8.2	11.7	12.6	11.4	10.9	9.6	10.0	9.9	8.6	9.5	8.7
Clear-Sky Solar Irradiance	taub	0.261	0.302	0.325	0.362	0.360	0.351	0.371	0.363	0.334	0.323	0.284	0.239	
	taud	2.219	2.313	2.396	2.370	2.436	2.483	2.442	2.458	2.527	2.472	2.304	2.137	
	Ebn at Noon	174	227	259	267	276	280	271	264	255	219	164	134	
	Edh at Noon	13	21	26	32	32	31	32	29	23	18	12	10	
All-Sky Solar Radiation	RadAvg	96	278	771	1211	1577	1821	1626	1277	852	377	126	65	
	RadStd	17	55	78	161	152	109	203	129	102	70	30	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+3.82	+4.57	N/A	N/A	N/A	-248	N/A	N/A	N/A
Regional (67 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	-236	N/A	N/A	N/A

Nomenclature: See separate page